

Biology Exit Exam 2018

1. Which of the following structure in arthropods contributed for the success of insects on earth
 - A. Air passes to cells through body fluid as in other higher animals
 - B. A versatile endoskeleton that allows flexibility and mobility
 - C. Metamorphosis in insects causes high intraspecific competition
 - D. **Segmentation and appendages provide for more efficient locomotion**
2. If photosynthesis is NOT takes place, what would happen for cells?
 - A. **Cellular respiration will no more continues after some time**
 - B. There will large accumulation of energy
 - C. Chloroplasts will become disappear
 - D. Cells become immediately die
3. Among the following biomolecules, which one is polymer?
 - A. Nucleotides
 - B. Monosaccharaides
 - C. **Nucleic acids**
 - D. Amino acids
4. During scientific report, under which section you want to give credit for those individuals helped you during research?
 - A. **Acknowledgement**
 - B. Statement of the problem
 - C. Literature review
 - D. Background
5. Which one of the following is a common and legitimate practice during germination to break seed dormancy?
 - A. Exposing seed to high temperature to break dormancy
 - B. **Soaking seeds in water for a day to allow water to enter into seed through micropile to break dormancy**
 - C. Putting seeds in salt water for some days
 - D. Placing water soaked seeds in refrigerator at freezing point to initiate germination
6. Order Isoptera are characterized by:
 - A. **Having similar wings size**
 - B. Having similar thorax and abdomen size
 - C. Having different leg size
 - D. Living inside water
7. Which of the following organisms have prokaryotic cells?
 - A. Fungi
 - B. Mammals
 - C. Birds
 - D. Bacteria
8. Which of the following is NOT related to ecological principles?
 - A. Organisms interact to their environment
 - B. Environmental factors influence the distribution of organisms in space and time

- C. Survival of species is determined by reproductive capability of a species and resource availability
- D. Most organisms can live independently in isolation in natural world
9. Identify false statement about the ecosystem functionality and dynamicity.
- A. Ecosystems are not permanent entities, rather show shifting pattern on the landscape
- B. Ecosystems are usually affected by natural and anthropogenic activities
- C. Biotic components are adapted to the environmental changes and always survive
- D. Energy and matter are continuously enter and exchange between ecosystems
10. What does the transmission of Ascaris takes place?
- A. Ingestion of contaminated food or water with parasite eggs or cysts
- B. Person-to-person contact or sexual activity
- C. Transmission through mosquito bite
- D. Larvae penetrate the skin, often from contaminated soil or water
11. Which of the following insects can be trapped using a sweep net?
- Question 10Answer

- a.
Termites
- b.
Butter flies
- c.
Ants
- d.
Mosquitoes

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b. Butterflies

Explanation: Sweep nets are used to catch flying and foliage-dwelling insects like butterflies, grasshoppers, and beetles. Termites and ants are generally collected from nests or the ground; mosquitoes are usually sampled with light traps, aspirators, or CO2 traps.

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Which of the following is true about skeletal muscles?

Question 11Answer

- a.
The muscle has non-striated, uni-nucleated cells
- b.
It is packed with branching chains of cells
- c.
It is situated in the walls of hollow visceral organs and blood vessels
- d.
It operates under voluntary control

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- d. It operates under voluntary control

Explanation: Skeletal muscle fibers are long, multinucleated, striated cells under conscious (voluntary) control. The other options describe smooth (a and c) or cardiac (b) muscle characteristics.

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In which groups of algae can you find all multicellular organisms?
Question 12Answer

- a.
Rhodophyta-red algae
- b.
Cyanopnyta-blue green algae
- c.
Bacilariophyta-Diatoms
- d.
Phaeophyta-brown algae

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d. Phaeophyta — brown algae

Explanation: Brown algae (Phaeophyta) include many large multicellular seaweeds (e.g., kelps). While Rhodophyta (red algae) also contain many multicellular species, the question asks for the group in which you can find all (or predominantly) multicellular organisms — Phaeophyta are primarily multicellular. Cyanophyta (blue-green algae) are prokaryotic and mostly unicellular or filamentous, and Bacillariophyta (diatoms) are unicellular (mostly) or form simple colonies.

10:55 PM

How are mutations important?

Question 13Answer

- a.
Mutations in homeotic genes are always lethal and have a minimal effect on evolution of animals
- b.
Mutations reduce genetic diversity of a population by reducing its allele frequency
- c.
Mutations reduce reproductive success of organisms such as antibiotic resistant bacteria.
- d.
Mutations produce new alleles or variant genes in a population that contribute for evolution.

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d. Mutations produce new alleles or variant genes in a population that contribute for evolution.

Explanation: Mutations are the source of genetic variation—creating new alleles that natural selection and other evolutionary forces can act on. The other options are incorrect: (a) some homeotic mutations are lethal but many are not and they can drive major evolutionary changes; (b) mutations generally increase genetic variation rather than reduce it; (c) mutations can increase reproductive success in some contexts (e.g., antibiotic resistance), not necessarily reduce it.

10:55 PM

Which one of the following is NOT protozoan disease?

Question 14Answer

- a.
Ascariasis
- b.
Giardiasis
- c.
Ameobiasis
- d.
Malaria

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- a. Ascariasis

Explanation: Ascariasis is caused by the parasitic roundworm *Ascaris* (a helminth), not a protozoan. Giardiasis (*Giardia*), Amebiasis (*Entamoeba*), and Malaria (*Plasmodium*) are protozoan diseases.

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During membrane transport, which molecules act as membrane pores?

Question 15Answer

- a.
Channel proteins
- b.
Carrier proteins
- c.
Gradient proteins

d.
Pump proteins

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a. Channel proteins

Explanation: Channel proteins form aqueous pores across the membrane that allow specific ions or small molecules to diffuse down their concentration or electrochemical gradients. Carrier proteins bind and transport molecules with conformational changes, and pump proteins use energy to move substances against gradients. "Gradient proteins" is not a standard term.

10:57 PM

In which of the following metabolisms does fatty acids are broken down into Acetyl-CoA units, entering the Krebs cycle?

Question 16Answer

a.
Nucleic acids Metabolism

b.
Lipid metabolism

c.
Carbohydrate metabolism

d.
Protein metabolism

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b. Lipid metabolism

Explanation: Fatty acids are broken down by beta-oxidation into two-carbon acetyl-CoA units, which then enter the Krebs (citric acid) cycle for energy production.

10:59 PM

Among the factors which one affects host-parasite relationship by altering the medium or environment for reproduction of microorganisms?

Question 17Answer

- a.
Immunity of the host
- b.
Virulence of the microorganism
- c.
Age of the host
- d.
pH

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d. pH

Explanation: pH alters the environmental conditions (medium) for microorganisms, affecting their survival, growth, and reproduction. The other factors (host immunity, pathogen virulence, host age) influence host–parasite interactions through biological or physiological mechanisms rather than directly changing the external medium for microbial reproduction.

11:00 PM

After plant specimens are collected from the field, what are the herbarium techniques used to keep specimens for future reference?

Question 18Answer

- a.
Herbarium does not has any procedural techniques or steps

- b.
Drying, labeling and gluing
- c.
Drying, grinding leaves and labeling
- d.
Keeping fresh leaves and covering with clean cloth

11:01 PM



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- b. Drying, labeling and gluing

Explanation: Standard herbarium techniques involve pressing and drying plant specimens, properly labeling them with collection data, and mounting (gluing or sewing) them onto herbarium sheets for long-term preservation and reference. Options c and d are incorrect because grinding or keeping fresh leaves are not appropriate for herbarium storage, and a is false.

11:01 PM

Which of the following structure in arthropods contributed for the success of insects on earth
Question 19Answer

- a.
Air passes to cells through body fluid as in other higher animals
- b.
A versatile endoskeleton that allows flexibility and mobility
- c.
Metamorphosis in insects causes high intraspecific competition
- d.
Segmentation and appendages provide for more efficient locomotion

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d. Segmentation and appendages provide for more efficient locomotion

Explanation: Arthropods (including insects) have segmented bodies and jointed appendages that allow specialization of body regions and efficient, versatile movement — a major factor in their evolutionary success. The other options are incorrect: insects use a tracheal system (not air through body fluids), they have an exoskeleton (not an endoskeleton), and metamorphosis typically reduces intraspecific competition by separating life-stage niches.

11:01 PM

Identify false statement about community characters.

Question 20Answer

a.

The flow of energy and materials is determined by different trophic structures

b.

It contains both biological and non-biological components

c.

It has diversified species in different growth form/stage/ and structures

d.

Some species might be dominant in the area

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b. It contains both biological and non-biological components

Explanation: A biological community specifically refers to the interacting populations of different species (biotic components). Non-biological (abiotic) components belong to the ecosystem concept. The other statements are true for communities: energy/material flow depends on trophic structure (a), communities have species with diverse growth forms and life stages (c), and some species may be dominant (d).

11:04 PM

If cells are starved and demanding high energy, what events possibly would takes place soon?
Question 21Answer

- a.
Mitochondria will not function
- b.
Photosynthesis will stop soon
- c.
Cellular respiration will stop soon
- d.
Glucose is directly converted in to pyruvate

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d. Glucose is directly converted into pyruvate

Explanation: When cells are starved and need rapid ATP, they increase glycolysis — the pathway that converts glucose to pyruvate — to produce ATP quickly (and to supply substrates for further ATP production via respiration or fermentation if oxygen is limited). The other options are incorrect:

- a) Mitochondria will still function; they typically increase activity to produce more ATP if substrates and oxygen are available.
- b) Photosynthesis is unrelated in most animal cells; in plants it isn't an immediate response to cellular energy demand.
- c) Cellular respiration would not stop; it usually increases to meet higher energy demand (unless substrates or oxygen are exhausted).

11:05 PM

The phenotypic gene expression may be affected by the following factors except:

Question 22Answer

- a.
Light intensity

b.
The reproductive rate

c.
Stress

d.
Temperature

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b. The reproductive rate

Explanation: Phenotypic gene expression is influenced by environmental factors such as light intensity, stress, and temperature, which can change gene regulation and lead to different phenotypes. Reproductive rate is a demographic/fitness parameter and does not directly affect how genes are expressed in an individual, so it is the exception.

11:06 PM

Which of the following is an environmental factor that determines host-parasite relationship?
Question 23Answer

a.
Age, behavior, immunity, nutrition, genetic makeup

b.
Immunity, virulence, lifecycle,

c.
Virulence, lifecycle, microbial density

d.
pH, temperature, resource availability

11:06 PM



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d. pH, temperature, resource availability

Explanation: Environmental factors that influence host–parasite relationships include abiotic conditions like pH, temperature, and availability of resources, which affect parasite survival, reproduction, and transmission. The other options list host factors (a) or pathogen/biological factors (b and c).

11:06 PM

Which of the following is an environmental factor that determines host-parasite relationship?
Question 23Answer

a.
Age, behavior, immunity, nutrition, genetic makeup

b.
Immunity, virulence, lifecycle,

c.
Virulence, lifecycle, microbial density

d.
pH, temperature, resource availability

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d. pH, temperature, resource availability

Explanation: Those are abiotic environmental factors that directly affect parasite survival, reproduction, transmission, and thus the host–parasite relationship. The other options list host-related or pathogen-related biological factors, not environmental ones.

11:07 PM

If you are conducting ecological studies and measuring the northing, easting and altitude, the type of data collection techniques or tools are:

Question 24Answer

- a.
Plot methods
- b.
GIS basis
- c.
Quadrat methods
- d.
Transect methods

11:08 PM



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b. GIS based

Explanation: Recording northing, easting, and altitude are geographic coordinates and elevation data—collected and managed using GIS (Geographic Information Systems) tools and GPS devices. Plot, quadrat, and transect methods are sampling techniques but do not inherently capture coordinate/elevation data unless combined with GIS.

11:08 PM

Which branch of biotechnology is mainly focuses on biofuel, food and beverage production?
Question 25Answer

- a.
Agricultural biotechnology
- b.
Medical biotechnology
- c.
Environmental biotechnology
- d.
Industrial biotechnology

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d. Industrial biotechnology

Explanation: Industrial biotechnology (also called white biotechnology) applies biological systems and organisms for large-scale industrial processes such as biofuel production, food and beverage fermentation, enzymes for manufacturing, and bioprocessing. Agricultural biotechnology focuses on crops and livestock improvement, medical biotechnology on healthcare and therapeutics, and environmental biotechnology on waste treatment and pollution control.

11:08 PM

Which branch of biotechnology is mainly focuses on biofuel, food and beverage production?

Question 25Answer

- a.
Agricultural biotechnology
- b.
Medical biotechnology
- c.
Environmental biotechnology
- d.
Industrial biotechnology

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d. Industrial biotechnology

Explanation: Industrial (white) biotechnology applies biological processes at scale for products such as biofuels, fermented foods and beverages, industrial enzymes, and bioprocessing technologies.

11:08 PM

In which Ethiopian Rift Valley Lakes does the Lates niloticus (Nile perch) found?

Question 26Answer

- a.
Lake Abijata and Lake Shalla
- b.
Lake Abaya and Lake Chamo
- c.
Lake Hawassa and Lake Tana
- d.
Lake Zeway and Lake Langano

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Correct answer: b. Lake Abaya and Lake Chamo

Explanation: Lates niloticus (Nile perch) is present in several Ethiopian Rift Valley lakes, notably Lake Abaya and Lake Chamo. The other listed lake pairs (Abijata & Shalla, Hawassa & Tana, Zeway & Langano) are not the typical pair associated with Nile perch in Ethiopia.

11:09 PM

If you are going to prepare a microbial culture, which of the following order would you follow?

Question 27Answer

- a.
Media Selection, Inoculation, Incubation, Isolation
- b.
Incubation, Isolation, Inoculation, Media Selection
- c.
Inoculation, Incubation, Isolation, Media Selection

d.
Media Selection, Incubation, Inoculation, Isolation

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a. Media Selection, Inoculation, Incubation, Isolation

Explanation: First choose the appropriate culture media, then inoculate the sample onto/into the medium, incubate under suitable conditions to allow growth, and finally isolate pure colonies or organisms for further study.

11:09 PM

In which area less diversity and distribution of commercially important fishes are found?

Question 28Answer

a.
Awash River

b.
Blue Nile & Tekeze Rivers

c.
Lake Tana, Baro & Omo-Gibe

d.
Southern Rift Valley Lakes (Abaya & Chamo)

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a. Awash River

Explanation: The Awash River basin has relatively lower diversity and distribution of commercially important fish compared with larger, more diverse systems like the Blue Nile &

Tekeze, Lake Tana/Baro/Omo-Gibe, or the Southern Rift Valley lakes (Abaya & Chamo), which support richer and more commercially valuable fish assemblages.

11:09 PM

Environmental stewardship promotes various conservation activities including one of the followings.

Question 29Answer

- a.
Ensuring unrestricted access of water for industries around water resources
- b.
Restricting aquatic resource from any form of use
- c.
Protecting aquatic resources from pollution and loss
- d.
Improving income of fishermen at the cost of aquatic resources

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- c. Protecting aquatic resources from pollution and loss

Explanation: Environmental stewardship promotes sustainable use and protection of natural resources—this includes preventing pollution, conserving habitats, and maintaining ecosystem health. The other options either encourage unsustainable use or are impractical extremes.

11:09 PM

During laboratory activity works, if your microscope is broken, what would you do?

Question 30Answer

- a.
Report to the instructor
- b.
Keep silent and never report the lab activities

c.
Collect the broken materials and hide them

d.
Try to maintain them soon

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a. Report to the instructor

Explanation: You should inform the instructor or lab supervisor immediately so they can ensure safety, arrange repair or replacement, and document the incident. Attempting unauthorized repairs, hiding damage, or staying silent can create safety hazards and disrupt the lab.

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During laboratory activity works, if your microscope is broken, what would you do?
Question 30Answer

a.
Report to the instructor

b.
Keep silent and never report the lab activities

c.
Collect the broken materials and hide them

d.
Try to maintain them soon

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a. Report to the instructor

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Consider observing cells in a stained sample using a light microscope. Which of the following activity should be done first?

Question 31Answer

- a.
Place a slide on the stage of a microscope
- b.
Set the microscope on a table and plug the power cord to a power outlet
- c.
Move and adjust the stage using adjusting knob
- d.
Select a highest magnification objective lens

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[Assistant](#)

- b. Set the microscope on a table and plug the power cord to a power outlet

Explanation: Before handling slides or changing knobs/ objectives, you should set the microscope on a stable surface and connect power (if using an electric light source). After that you place the slide, start with the lowest-power objective and then use the stage knobs to focus and move to higher magnifications.

11:13 PM

What is the probability of getting the genotype aabb offspring from a dihybrid cross between two parent plants with the genotypes of AaBb and AaBb?

Question 32Answer

- a.
 $\frac{1}{8}$
- b.
 $\frac{1}{4}$

c.
1/16

d.
1/2

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Correct answer: c. 1/16

Explanation: For each gene locus, two Aa parents produce offspring genotypes with probabilities: AA 1/4, Aa 1/2, aa 1/4. The probability of aa from $Aa \times Aa$ is 1/4; similarly bb from $Bb \times Bb$ is 1/4. For independent genes, multiply the probabilities: $(1/4) \times (1/4) = 1/16$.

11:13 PM

What are environmental factors mainly affecting microbial growths?

Question 33Answer

a.
Secondary metabolites

b.
Rainfall

c.
Infectious diseases

d.
Nutrients

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d. Nutrients

Explanation: Nutrients (carbon, nitrogen, minerals, etc.) are primary environmental factors that directly determine microbial growth rates and biomass. Other important environmental factors include temperature, pH, oxygen availability, and moisture. The other options are not primary environmental determinants of microbial growth (secondary metabolites are products, rainfall is an indirect factor, and infectious diseases are biological interactions).

11:14 PM

Which one of the following is advantage of genetic mutation?

Question 34Answer

- a.
Change in chromosome number
- b.
Incorporate different amino acids in to the proteins
- c.
Allow organisms to evolve
- d.
Change in chromosome structure

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- c. Allow organisms to evolve

Explanation: Mutations introduce genetic variation (new alleles) that natural selection and other evolutionary forces can act on, enabling evolution. The other options describe types or effects of mutations (changes in chromosome number or structure, altered amino acids) but are not advantages in themselves.

11:14 PM

Which of the following statistical tests is best for comparison of means of three independent samples collected from different populations?

Question 35Answer

- a.
Analysis of variance

b.
Student's t-test

c.
Correlation

d.
Chi-square test

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a. Analysis of variance

Explanation: ANOVA (Analysis of Variance) is used to compare the means of three or more independent groups to determine if at least one group mean differs significantly from the others. Student's t-test compares two groups; correlation assesses association between variables; chi-square tests categorical data.

11:14 PM

When you are using insect collecting tools like sweep nets, what procedures should be followed?
Question 36Answer

a.
Collecting using hand is more easier than using seep nets

b.
The sweep nets hole size must be enough to escape insects

c.
Insect skeletons should be digested very well

d.
Insect parts and bodies must be properly handled

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d. Insect parts and bodies must be properly handled

Explanation: When using sweep nets and other collecting tools, specimens should be handled carefully to avoid damage, preserve diagnostic features, and ensure proper labeling and storage. This includes gentle transfer to killing jars or vials, careful pinning/mounting, and using appropriate preservation methods. The other options are incorrect or unsafe.

11:15 PM

Suppose a pandemic disease is occurred and stayed for long period of time. As biotechnologist scholar, what would you do?

Question 37Answer

- a.
Inform community to be exposed and resist the disease
- b.
Recommend the community to stay at home
- c.
Use microorganisms to develop vaccine
- d.
Biotechnologists are free of such issues

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c. Use microorganisms to develop vaccine

Explanation: As a biotechnologist, an appropriate response during a prolonged pandemic is to apply biotechnology tools—such as using microorganisms (or their components) to develop vaccines, produce diagnostic reagents, and create therapeutic biologics. Options b (recommend stay-at-home) is a public-health measure but not specific to the role of a biotechnologist; a is unsafe and incorrect; d is false — biotechnologists play a central role in responding to pandemics.

11:16 PM

Which of the following is true about glycolysis?

Question 38Answer

- a.
During glycolysis, a single glucose is split into two three-carbon sugars
- b.
It is anabolic reaction and the reaction is not catalyzed by enzymes
- c.
Glycolysis takes place in the mitochondria of cells
- d.
It produces 2 NAD⁺ and net 4ATP

11:16 PM



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- a. During glycolysis, a single glucose is split into two three-carbon sugars

Explanation: Glycolysis breaks one 6-carbon glucose into two 3-carbon pyruvate molecules (via 3-carbon intermediates). Other options are incorrect:

- b) Glycolysis is a catabolic pathway and is enzyme-catalyzed.
- c) Glycolysis occurs in the cytoplasm, not in mitochondria.
- d) Glycolysis produces 2 NADH (not NAD⁺ net) and a net gain of 2 ATP (not 4).

11:16 PM

Which one of the following is NOT the role of adaptive immunity?

Question 39Answer

- a.
It forms memory of antigens from past infections allowing faster and stronger response upon re-exposure
- b.
It is a first line of defense that protect the body against general threats

c.

Mediated by T cells, directly kills infected cells or regulating the immune response

d.

Mediated by B cells, it attacks invaders by producing specific antibodies

11:17 PM



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b. It is a first line of defense that protect the body against general threats

Explanation: Adaptive immunity is specific and slower to activate on first exposure; it provides antigen-specific responses, immunological memory, B-cell antibody production, and T-cell-mediated killing/regulation. The innate immune system (not adaptive immunity) serves as the first line of defense against general threats.

11:17 PM

Why do you use quadrats?

Question 40Answer

a.

To characterize plants as monocot and dicots

b.

To study feeding relationship of organisms in an ecosystem

c.

To count individuals of a species in an area

d.

To gather data required to calculate density of species

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d. To gather data required to calculate density of species

Explanation: Quadrats are sampling frames used in ecological studies to estimate abundance, density, frequency, and distribution of organisms within a defined area. While they can also be used to count individuals (c), the primary purpose framed here is to collect data needed for density calculations.

11:17 PM

Which of the following technique is NOT a reliable way to detect parasites in stools containing low concentration of parasites?

Question 41Answer

- a.
Direct microscopic examination of fecal smears.
- b.
Sedimentation, centrifugation, flotation using brine,
- c.
Quantitative methods include the filtration-count technique and Stoll's dilution-count technique.
- d.
Centrifugal flotation using zinc sulphate solution and the acid-ether techniques

11:18 PM



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a. Direct microscopic examination of fecal smears.

Explanation: Direct smears are quick but have low sensitivity for samples with low parasite concentration. Concentration techniques (sedimentation, flotation, centrifugal flotation with zinc sulfate or acid-ether, and quantitative dilution/filtration-count methods) increase the likelihood of detecting parasites present in low numbers.

11:18 PM

To which class does an arthropod that has dorsoventrally flattened body; variable number of segments, each with one pair of legs; one pair of long antennae; and oviparous, belongs?

Question 42Answer

a.
Diplopoda

b.
Chilopoda

c.
Insecta

d.
Symphyla

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b. Chilopoda

Explanation: Centipedes (class Chilopoda) have dorsoventrally flattened bodies with a variable number of segments and one pair of legs per segment, one pair of long antennae, and are oviparous. Millipedes (Diplopoda) have two pairs of legs per segment. Insects (Insecta) have three major body regions and three pairs of legs total. Symphyla are small, soil-dwelling myriapods with distinct characteristics and are less commonly described with those features.

11:19 PM

Which microorganisms are found in the boarder edge of living and non-living things?
Question 43Answer

a.
Bacteria

b.
Viruses

c.
Protozoans

d.
Fungus

11:21 PM



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d. Fungus

Explanation: Fungi (and fungal-like organisms) commonly colonize the interface between living and non-living materials — for example, decaying plant or animal matter, building materials, and stored foods. They thrive on dead organic substrates while often interacting with living hosts (as decomposers, saprophytes, or pathogens). Bacteria and protozoans also inhabit such interfaces but fungi are especially characteristic for visibly colonizing the boundary of living and non-living matter (e.g., mold on food, wood decay). Viruses are obligate intracellular parasites and are not found growing on non-living surfaces in the same way.

11:21 PM

One of the following is likely excluded from sense organs.

Question 44Answer

- a.
The sense of taste
- b.
The sense of hearing
- c.
The sense of sight
- d.
The sense touch

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a. The sense of taste

Explanation: In many classifications of sense organs, taste is often grouped with the chemical senses (gustation and olfaction) rather than with the classic sense organs (eye for sight, ear for

hearing, skin for touch). If the question intends the classic external sense organs, taste (which depends on taste buds inside the mouth) is the one typically excluded.

11:22 PM

Which of the following organ system is correctly associated with its primary function in our body?

Question 45Answer

- a.
Endocrine system involves electrical impulses in response to stimuli passes through sensory organs
- b.
Digestive system involves breakdown of larger molecules in foods into smaller parts
- c.
Respiratory system involves the production of larger molecules from smaller particles
- d.
Nervous system involves response by some glands in the body triggered by secretion of chemical stimuli

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- b. Digestive system involves breakdown of larger molecules in foods into smaller parts

Explanation: The digestive system's primary function is to mechanically and chemically break down food into smaller molecules (e.g., proteins → amino acids, carbohydrates → sugars, fats → fatty acids) that can be absorbed. The other options are incorrect/misstated: (a) describes nervous system electrical impulses, not endocrine; (c) describes anabolic processes (e.g., biosynthesis) not the respiratory system; (d) mixes nervous and endocrine functions—nervous system uses electrical signals, while glands respond to chemical (hormonal) signals from the endocrine system.

11:22 PM

Which of the following process has its origin on traditions biotechnology approaches?

Question 46Answer

- a.
Polymerase chain reaction used in genetic or DNA fingerprinting
- b.
Extraction of DNA from plants
- c.
Production of genetically modified crops
- d.
Production of industrial alcohol by fermentation

11:23 PM



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d. Production of industrial alcohol by fermentation

Explanation: Traditional (or classical) biotechnology includes age-old practices that use living organisms or their enzymes for useful processes—fermentation to produce alcohol, bread, cheese, and other products is a prime example. The other options are modern molecular biotechnology techniques:

- a. PCR is a modern molecular method developed in the 1980s.
- b. Extraction of DNA is a laboratory molecular biology procedure (modern technique).
- c. Production of genetically modified crops is a product of recombinant DNA/genetic engineering (modern biotechnology).

11:23 PM

A prolonged dormancy (arrest of growth) in many insects regardless of environment, whether or not favorable conditions prevail is called?

Question 47Answer

- a.
Diapause
- b.
Molting
- c.
Metamorphosis

d.
Migration

11:23 PM



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a. Diapause

Explanation: Diapause is a hormonally controlled, prolonged arrest of development in insects that can occur regardless of current environmental conditions, allowing survival through unfavorable seasons. Molting is shedding the exoskeleton; metamorphosis is developmental change between life stages; migration is seasonal movement.

11:23 PM

Economically beneficial and social insects are found in which order?
Question 48Answer

a.
Lepidoptera

b.
Isopteran

c.
Diptera

d.
Hymenoptera

11:23 PM



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d. Hymenoptera

Explanation: Hymenoptera (bees, wasps, ants) includes many socially organized insects (e.g., honeybees, some ants, social wasps) that provide major economic and social benefits—pollination, honey production, wax, biological pest control, and ecological services. The other orders include important insects too (Lepidoptera—butterflies/moths; Diptera—flies; Isoptera/Isoteran—termites), but Hymenoptera are most notable for their economic and social roles.

11:23 PM

Identify true statement about the vascular tissue organization of higher plants.

Question 49Answer

a.

Xylem is made up of vessel and sieve cells

b.

Phloem is made up of sieve and vessel cells

c.

Both xylem and phloem are made up of tracheid and companion cells

d.

Xylem is made up of tracheid and vessel cells

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d. Xylem is made up of tracheid and vessel cells

Explanation: In vascular plants, xylem contains tracheids and vessel elements (vessel cells) that conduct water and provide support. Phloem is made up of sieve elements (sieve tube elements in angiosperms) and companion cells, not vessels or tracheids.

11:24 PM

Which of the following groups of eukaryotic plants is water living and its reproduction is totally dependent on water?

Question 50Answer

a.

Bryophytes

b.
Pteridophytes

c.
Embryophyta

d.
Algae

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[Assistant](#)

a. Bryophytes

Explanation: Bryophytes (mosses, liverworts, hornworts) are non-vascular plants that require water for the motile sperm to swim to the egg during sexual reproduction, so their reproduction is highly dependent on water. Pteridophytes (ferns and allies) also need water for sperm movement but are vascular; algae are a diverse group (some are aquatic, some terrestrial) and not all are eukaryotic "plants" in the strict sense; Embryophyta is the land-plant clade (includes bryophytes and vascular plants) and not all members depend entirely on water for reproduction.

11:24 PM

The following are contributions of wetlands to wellbeing of animals except:

Question 51Answer

a.
Sediment retention and accumulation organic matter

b.
Habitat for breeding site

c.
Generation of O₂ and sink CO₂

d.
Harbor polluted water and toxic elements

11:24 PM



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d. Harbor polluted water and toxic elements

Explanation: Wetlands provide many ecosystem services beneficial to animals and humans—retaining sediments and accumulating organic matter (a), serving as breeding habitat (b), and producing oxygen while sequestering carbon dioxide (c). While wetlands can sometimes accumulate pollutants, "harboring polluted water and toxic elements" is not a beneficial contribution; rather, it is a negative impact or risk.

11:24 PM

How do enzymes Work?

Question 52Answer

a.

Enzymes increase temperature and pH of medium where chemical reaction takes place

b.

Enzymes react with substrates to form new product

c.

Enzymes change permanently after a chemical reaction is completed

d.

Enzymes reduce activation energy required to start a chemical reaction

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d. Enzymes reduce activation energy required to start a chemical reaction

Explanation: Enzymes are biological catalysts that lower the activation energy for reactions, increasing reaction rates without being consumed or permanently changed. They bind substrates to form enzyme–substrate complexes (but do not become permanent parts of the products), and their activity is sensitive to temperature and pH (but they do not increase them).

11:25 PM

How environmental and aquatic resources are respected?

Question 53Answer

- a.
Use them wisely, sustainably & do conservation
- b.
Feel their contribution is minor & insignificant
- c.
Feel they are wastelands
- d.
They are not good resources since they are watery

11:25 PM



[Assistant](#)

- a. Use them wisely, sustainably & do conservation

Explanation: Respecting environmental and aquatic resources means using them sustainably, conserving habitats, preventing pollution, and managing resources so they continue to provide ecosystem services for current and future generations. The other options reflect disregard or misinformation.

11:25 PM

The collections of different alleles is termed as:

Question 54Answer

- a.
Inheritance
- b.
Genotype
- c.
Genes

d.
Chromosomes

11:26 PM



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b. Genotype

Explanation: A genotype is the set (collection) of alleles an organism carries for one or more genes. "Inheritance" describes transmission of genes across generations, "genes" are units of heredity themselves, and "chromosomes" are structures that carry many genes.

11:26 PM

One of the following is inappropriate conduct while dealing with laboratory activities and conducting research?

Question 55Answer

a.
Develop a simple and general safety plans or protocols which is similar for all experiments that have potential risks

b.
Maintain good conduct while working with others in the lab

c.
Be honest with data usage and respect patient confidentiality in clinical and other relevant settings

d.
Make sure that the materials you are working with are clean enough to prevent skewed results or chance of infections

View more

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a. Develop a simple and general safety plans or protocols which is similar for all experiments that have potential risks

Explanation: This is inappropriate because safety plans must be specific to each experiment's hazards and risks; a one-size-fits-all protocol can miss critical safety measures. The other options describe appropriate conduct: maintaining good teamwork, honest data handling and confidentiality, and proper cleanliness of materials.

11:27 PM

Which of the following metabolic pathways produces the highest ATP?

Question 56Answer

a.
Oxidative Phosphorylation

b.
Oxidative Decarboxylation

c.
Krebs Cycle

d.
Glycolysis

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[Assistant](#)

a. Oxidative Phosphorylation

Explanation: Oxidative phosphorylation (the electron transport chain plus ATP synthase in the inner mitochondrial membrane) generates the majority of ATP produced during aerobic respiration — typically ~26–34 ATP per glucose — by using the proton gradient established by electron transport. The Krebs cycle and oxidative decarboxylation yield reduced cofactors (NADH, FADH₂) and a small direct amount of ATP/GTP, and glycolysis yields a net 2 ATP per glucose.

11:27 PM

What does the Polymerase Chain Reaction (PCR) in molecular biology do?

Question 57Answer

- a.
Joins two segments of DNA
- b.
Slices DNA segments from long chain
- c.
Isolate DNA from other cell contents
- d.
Amplifies specific DNA sequences

11:27 PM



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- d. Amplifies specific DNA sequences

Explanation: PCR (polymerase chain reaction) is a laboratory technique that makes many copies of a specific DNA segment, allowing detection, cloning, sequencing, or further analysis of that target sequence.

11:27 PM

Among important biomolecules, which are best efficient energy sources?
Question 58Answer

- a.
Vitamins
- b.
Carbohydrates
- c.
Proteins
- d.
Lipids

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d. Lipids

Explanation: Lipids (fats) yield more than twice the energy per gram (~9 kcal/g) compared with carbohydrates or proteins (~4 kcal/g), making them the most efficient long-term energy store. Vitamins are not energy sources.

11:28 PM

What are the fundamental units of the Nervous system?

Question 59Answer

a.

Axon

b.

Cell body

c.

Neurons

d.

Dendrites

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c. Neurons

Explanation: Neurons are the fundamental functional units of the nervous system; they consist of a cell body (soma), dendrites, and an axon, and they transmit electrical and chemical signals.

11:28 PM

Which class of Immunoglobulin (Ig) is associated with receptor and functions on B-lymphocyte cell membranes?

Question 60Answer

a.
IgE

b.
IgM

c.
IgD

d.
IgG

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c. IgD

Explanation: IgD is primarily found as a membrane-bound immunoglobulin on naive B-lymphocytes where it functions as an antigen receptor. (IgM is also present on immature B cells as a surface receptor; IgG and IgE are mainly secreted antibodies.)

11:28 PM

Which group of gymnosperm has the largest and most commercially important members that are known for their cold climate adaptations and use as timber?

Question 61Answer

a.
Coniferophyta

b.
Cycadophyta

c.
Ginkgophyta

d.
Gnetophyta

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[Assistant](#)

a. Coniferophyta

Explanation: Conifers (Coniferophyta) include the largest and most commercially important gymnosperms—pines, firs, spruces, cedars, and redwoods—well adapted to cold climates and widely used for timber, paper, and resin. The other gymnosperm groups (Cycadophyta, Ginkgophyta, Gnetophyta) have far fewer species and less commercial timber significance.

11:30 PM

In which molecular mechanism does the nitrogen bases in DNA molecule are disposed to detect and copy the genetic information to synthesize an individual gene or genes takes place?

Question 62Answer

a.
Transformation

b.
Transcription

c.
Translation

d.
Replication

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b. Transcription

Explanation: Transcription is the molecular process in which the sequence of nitrogenous bases in a DNA segment (a gene) is read and copied into a complementary RNA molecule (usually messenger RNA). Translation is the subsequent step where the mRNA sequence is used to synthesize a protein. Replication copies the entire DNA for cell division, and transformation is uptake of external DNA by a cell.

11:30 PM

Which of the following alternatives correctly presents the development of microbiology and its application?

Question 63Answer

a.

The development of microbiology as a science has allowed us to control microbes and use them for our benefit

b.

Environmental microbiology focuses on preventing and treating infectious diseases

c.

Agricultural microbiology deals with industrial application of microorganisms, such as fermented beverages and pharmaceuticals

d.

Medical microbiology deals about the ecology and distributed of microorganisms on the earth and its atmosphere

View more

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[Assistant](#)

a. The development of microbiology as a science has allowed us to control microbes and use them for our benefit

Explanation: Microbiology advances have enabled disease control (vaccines, antibiotics), food safety and fermentation, industrial biotechnology, and environmental applications. The other options misstate specialties: environmental microbiology studies ecology and roles of microbes in environments; agricultural microbiology deals with plant/soil/microbe interactions and crop health; medical microbiology focuses on infectious diseases, their prevention and treatment.

11:31 PM

What do you think that there are certain bioethical issues related with cloning?

Question 64Answer

a.

It is because of cloning is fast events

b.

It has no any more problems, rather limited awareness about cloning

c.

It is because of economical and producing large copies

d.

It is because of cloning is producing identical cop

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d. It is because cloning is producing identical copies

Explanation: A core bioethical concern about cloning is that it produces genetically identical organisms, raising issues about individuality, identity, consent (for humans), biodiversity reduction, potential harm to clones, and social and moral implications of replicating organisms. The other options do not accurately capture the primary ethical concerns.

11:31 PM

Which of the following methods for studying cells is the simplest, most common, inexpensive, and uses visible light to view living cells, observe cell movement, and general cell structure?

Question 65Answer

a.

Biochemical Techniques

b.

Optical Microscopy

c.

Cell Culture

d.

Molecular techniques

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b. Optical Microscopy

Explanation: Optical (light) microscopy is the simplest, most common, and inexpensive method that uses visible light to observe living cells, their movements, and general structures.

11:32 PM

How does the recombinant DNA technology in genetic engineering allows production of desired protein or trait?

Question 66Answer

a.

By modifying an organism's genetic makeup to achieve specific outcomes

b.

By selecting highly successful trait and encouraging it to mutate

c.

By regulating environmental condition of organisms

d.

By performing hybridization of alleles to get pure progeny

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[Assistant](#)

a. By modifying an organism's genetic makeup to achieve specific outcomes

Explanation: Recombinant DNA technology creates new combinations of DNA by inserting genes (that encode desired proteins or traits) into host organisms. The host then expresses the introduced gene, producing the protein or showing the trait. The other options describe selective breeding, environmental manipulation, or hybridization, which are different methods.

11:32 PM

If certain types of crops are frequently affected by pest, what would be your suggestion as a molecular Biologist?

Question 67Answer

- a.
Use pesticides that can kill pests
- b.
Applying crop rotation techniques
- c.
Use recombinant DNA technology to introduce resistant genes
- d.
Grow crops mixing with others

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c. Use recombinant DNA technology to introduce resistant genes

Explanation: As a molecular biologist, recommending the introduction of pest-resistance genes (via recombinant DNA/genetic engineering) is appropriate because it can provide durable, targeted protection against pests while reducing reliance on chemical pesticides. Engineered resistance can be designed for specificity, reduce crop losses, and lower environmental and human health risks from broad-spectrum pesticides.

Notes/considerations:

- Integrated Pest Management (IPM) is best practice: combine genetic resistance with cultural methods (crop rotation, intercropping), biological control, and judicious pesticide use when needed.
- Evaluate biosafety, regulatory approval, potential non-target effects, resistance management (to delay pest adaptation), and socioeconomic factors before deployment.

11:32 PM

Which one of the following has NO direct role in the physiological mechanisms of homeostasis?

Question 68Answer

- a.
Receptors
- b.
Effectors
- c.
The environment
- d.
Control centers

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[Assistant](#)

Correct answer: c. The environment

Explanation: Homeostasis is maintained by receptors (which detect changes), control centers (which process information and set setpoints), and effectors (which enact responses to restore balance). The environment can influence internal conditions but is not a direct component of the physiological feedback mechanisms that maintain homeostasis.

11:33 PM

Which of the following practices does NOT require biotechnological application?

Question 69Answer

- a.
Biofertilizers and Biopesticides production
- b.
Breeding of local varieties
- c.
Crop improvement and nutritional enhancement
- d.
Vaccine production

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b. Breeding of local varieties

Explanation: Traditional breeding of local varieties can be carried out using conventional selection and mating techniques without biotechnological methods. The other options—biofertilizer/biopesticide production, crop improvement/nutritional enhancement (often involving genetic engineering or molecular tools), and vaccine production—typically require biotechnology.

11:33 PM

In eukaryotes, which of the following conditions of gene regulation occurs at the level of genome (DNA)?

Question 70 Answer

- a.
Transcriptional regulation by protein A24
- b.
Total genetic suspension or shutdown occurs
- c.
Modifications of histones or histone methylation
- d.
Modification of DNA sequences or DNA methylation

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d. Modification of DNA sequences or DNA methylation

Explanation: Regulation at the genome (DNA) level in eukaryotes includes changes to the DNA itself or its chemical modification—most notably DNA methylation—which affects gene accessibility and long-term transcriptional activity. Histone modifications (option c) act at the

chromatin level (also affecting accessibility) but are considered chromatin-level rather than direct DNA-level changes. Options a and b describe transcriptional or extreme/incorrect scenarios rather than genome-level regulation.

11:34 PM

What is the function of DNA?

Question 71Answer

- a.
Used for direct protein synthesis
- b.
Used for ribosome production
- c.
Used for direct disease protection
- d.
Used for storing the genetic information

11:34 PM



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- d. Used for storing the genetic information

Explanation: DNA's primary function is to store the hereditary information (the genetic code) that directs development, functioning, and reproduction of living organisms. It is transcribed into RNA and then translated into proteins; it does not directly synthesize proteins or produce ribosomes nor does it directly protect against diseases.

11:34 PM

What is the primary cause of eutrophication in fresh water bodies?

Question 72Answer

- a.
Removal of vegetation at shores of fresh water bodies

- b.
Accumulation of fertilizers and pesticides in fresh water bodies due to erosion of agricultural soils into fresh water bodies
- c.
Siltation in fresh water bodies caused by sheet erosion from degraded upper catchment areas
- d.
Release of industrial wastes into fresh water bodies

11:35 PM



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- b. Accumulation of fertilizers and pesticides in fresh water bodies due to erosion of agricultural soils into fresh water bodies

Explanation: Eutrophication is primarily caused by excess nutrients—especially nitrogen and phosphorus—entering water bodies, most commonly from fertilizer runoff and agricultural erosion. These nutrients stimulate algal blooms, which deplete oxygen when they die and decompose, harming aquatic life. While siltation, shoreline removal, and industrial wastes can degrade water quality, nutrient enrichment from agricultural runoff is the main driver of eutrophication.

11:35 PM

The following environmental factors have regulatory effect on the communities except:

Question 73Answer

- a.
Environmental heterogeneity
- b.
Predation
- c.
Competition
- d.
The gene pool

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d. The gene pool

Explanation: Environmental factors that regulate communities include abiotic heterogeneity (a), and biotic interactions like predation (b) and competition (c). The gene pool is a genetic property of populations (intraspecific), not an environmental regulator of community structure.

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Which enzyme class is responsible to catalyze reactions that break bonds by adding water?

Question 74Answer

a.

Isomerases

b.

Oxidoreductases

c.

Transferases

d.

Hydrolases

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d. Hydrolases

Explanation: Hydrolases catalyze hydrolysis reactions, breaking chemical bonds by the addition of water (e.g., proteases, lipases, nucleases). The other classes perform different reactions: oxidoreductases (redox), transferases (group transfer), and isomerases (intramolecular rearrangements).

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The ability of microorganism to cause a disease is termed as
Question 75Answer

- a.
Species identification
- b.
Immunity
- c.
Pathogenicity
- d.
Microbial control

11:36 PM



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c. Pathogenicity

Explanation: Pathogenicity is the ability of a microorganism to cause disease in a host.

11:36 PM

What do think about high diversity and significant ecological contributions of higher plants?
Question 76Answer

- a.
They restricted to temperate region
- b.
They have different ecological adaptations
- c.
They are free from any disturbances
- d.
They are selected by nature

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[Assistant](#)

b. They have different ecological adaptations

Explanation: High diversity and major ecological roles of higher plants are due to their wide range of ecological adaptations (morphological, physiological, reproductive) that allow them to occupy diverse habitats. The other options are incorrect: they are not restricted to temperate regions, they are subject to disturbances, and "selected by nature" is vague and not a specific explanation.

11:38 PM

Which of the following Medelian law states that the separation of two alleles for a single gene during gamete formation, so that each gamete receives only one allele?

Question 77 Answer

a.
Law of Independent Assortment

b.
Law of Segregation

c.
Law of Co-dominance

d.
Law of Dominance

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b. Law of Segregation

Explanation: Mendel's Law of Segregation states that the two alleles for a given gene separate during gamete formation, so each gamete receives only one allele.

11:39 PM

As anatomist, you want to identify regions of the heart and points on the chest where certain heart sounds can best be heard. How you can do this?

Question 78Answer

- a.
Searching stomach
- b.
Searching vertebrae
- c.
Find breastbone & parts of the ribs can be seen on the front of the chest
- d.
Searching spinal cords that cover heart

11:39 PM



[Assistant](#)

- c. Find breastbone & parts of the ribs can be seen on the front of the chest

Explanation: To auscultate heart sounds you locate surface landmarks on the anterior chest—sternum (breastbone) and ribs/intercostal spaces—and use those to identify cardiac auscultation points (e.g., aortic, pulmonic, tricuspid, and mitral areas).

11:39 PM

Suppose you are applying breeding experiments using Mendelian principles in laboratories, what are the correct steps to apply it?

Question 79Answer

- a.
Data analysis, select-true breeding parents (P), perform initial cross (PxF1) & produce F2 generation
- b.
Select-true breeding parents (P), perform initial cross (PxF1), produce F2 generation & data analysis

c.
Data analysis, produce F2 generation, perform initial cross (PxF1) & select-true breeding parents (P)

d.
Perform initial cross (PxF1), produce F2 generation, data analysis & select-true breeding parents (P)

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b. Select-true breeding parents (P), perform initial cross (P x) to produce F1, produce F2 generation & data analysis

Explanation: Mendelian breeding experiments start with choosing true-breeding (homozygous) parent strains (P), crossing them to produce the F1 generation, allowing F1 to self (or intercross) to produce F2, and then analyzing the offspring data to infer inheritance patterns.

11:40 PM

What do you think about the use of infectious materials and tools used for studying parasites?
Question 80Answer

a.
We have to handle them carefully and properly

b.
Materials linked with parasites are risk for health, and must be thrown at the end

c.
At the end of the work, avoid them

d.
Since they are contaminated with parasites, proper handling is not required

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[Assistant](#)

a. We have to handle them carefully and properly

Explanation: Infectious materials and tools used to study parasites can pose health risks and must be handled with appropriate biosafety practices—proper use of PPE, containment, decontamination, and disposal—to protect personnel and the environment. Options b–d are unsafe or incorrect.

11:40 PM

Which one of the following is true about innate and adaptive immune systems?

Question 81Answer

a.
Adaptive immune is specific and obtained before birth

b.
Innate immune is specific and obtained after birth

c.
Adaptive immune is specific and obtained after birth

d.
Innate immune is non-specific and obtained after birth

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c. Adaptive immune is specific and obtained after birth

Explanation: The adaptive immune system develops after exposure to antigens (after birth), is antigen-specific, and provides memory. The innate immune system is non-specific and is present at birth.

11:41 PM

If you want to identify microbes using microscopic techniques, what procedures must be sued orderly?

Question 82Answer

a.
Identification, isolation, incubation & inoculation

b.
Inoculation, incubation, isolation & identification

c.
Inoculation, isolation, incubation & identification

d.
Isolation, incubation, inoculation & identification

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[Assistant](#)

b. Inoculation, incubation, isolation & identification

Explanation: First inoculate the sample onto appropriate media, then incubate to allow growth, isolate pure colonies/organisms, and finally identify them using morphological, biochemical, molecular, or microscopic techniques.

11:41 PM

Identify how parasites are affecting hosts and cause diseases?
Question 83Answer

a.
They obtain most nutrients from the environment, but they need hosts only for reproduction

b.
Their primary environment is host, but eggs & juveniles can survive in water and soil

c.
They are always affecting the hosts

d.
They are usually live inside the hosts

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b. Their primary environment is host, but eggs & juveniles can survive in water and soil

Explanation: Many parasites spend most of their lifecycle in a host (where they feed, grow, and reproduce) but release eggs, larvae, or cysts into the environment (soil, water, or vectors) where those stages can survive until they infect a new host. The other options are incorrect: (a) parasites rely on hosts for nutrients as well as reproduction; (c) not all parasites always cause disease—some are relatively benign or cause disease only under certain conditions; (d) while many parasites live inside hosts (endoparasites), some live on the host surface (ectoparasites) or have free-living stages outside the host.

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Question 84

Not yet answered

Flag question

Question text

Mycobacteria and some actinomycetes such as *Micobacter tuberculosis* and *Micobacter leprae* contain a wax-like material in their cell wall. Which stain will help you observe these microorganisms?

Question 84Answer

a.

Negative Staining

b.

The Leifson Flagella Stain

c.

The Acid-Fast Stain

d.

The Gram Stain

View more

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c. The Acid-Fast Stain

Explanation: Mycobacteria (e.g., *Mycobacterium tuberculosis*, *M. leprae*) have mycolic acids (waxy lipids) in their cell walls that resist conventional stains; the acid-fast stain (e.g., Ziehl–Neelsen or Kinyoun) is used to visualize these organisms.

11:42 PM

What are some representative viral diseases?

Question 85Answer

a.

Common cold, influenza & ebola

b.

Taeniasis & schistosomiasis

c.

Amoebiasis, ascariasis

d.

Malaria & giardia

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a. Common cold, influenza & ebola

Explanation: These are viral diseases—common cold (many rhinoviruses and coronaviruses), influenza (influenza viruses), and Ebola (Ebolavirus). The other options list parasitic diseases (taeniasis, schistosomiasis, amoebiasis, ascariasis, giardiasis) or protozoan/helminth-borne illnesses (malaria).

11:44 PM

Which structures are likely found in viruses?

Question 86Answer

a.
RNA or DNA

b.
Ribosomes

c.
Cell wall

d.
Nucleus

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[Assistant](#)

a. RNA or DNA

Explanation: Viruses contain genetic material, which can be either RNA or DNA (single- or double-stranded). They lack cellular structures such as ribosomes, cell walls, and a nucleus.

11:45 PM

The following are techniques you could apply during water sample collection except:

Question 87 Answer

a.
Make ready standard sampling, storing, testing & analysis materials

b.
Collect composite water samples

c.
Collect water from one site

d.
Consider biological organisms, timing and frequency of collection

11:45 PM



[Assistant](#)

c. Collect water from one site

Explanation: Best practices for water sampling include preparing standard sampling/storing/testing materials, collecting composite samples (or multiple samples over time/space), and considering biological targets, timing, and sampling frequency. Collecting from only one site is usually insufficient to characterize water quality comprehensively.

11:45 PM

What feature distinguishes protozoans from helminthes and arthropods?

Question 88Answer

a.

Life Stages that consists of eggs, larvae, and mature adult stage

b.

Multicellular organization

c.

Large in size and can be seen with naked eye

d.

Reproduce quickly usually by asexually (binary fission)

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d. Reproduce quickly usually by asexually (binary fission)

Explanation: Protozoans are unicellular and commonly reproduce rapidly by asexual methods such as binary fission. In contrast, helminths and arthropods are multicellular, often have complex life stages (eggs, larvae, adults), and are large enough to be seen without a microscope.

11:45 PM

How quantitative researches differ from the qualitative research?

Question 89Answer

- a.
It uses interviews and observations
- b.
There is no formal measurement
- c.
It is an in-depth examination of one person's case study
- d.
It contains statistical analysis, formal and systematic measures

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- d. It contains statistical analysis, formal and systematic measures

Explanation: Quantitative research uses structured measurement, numeric data, and statistical analysis to test hypotheses and quantify relationships. The other options describe qualitative approaches (interviews/observations, in-depth case studies) or lack of formal measurement.

11:46 PM

What is homeostasis? Homeostasis is:
Question 90Answer

- a.
The body's attempt to maintain relatively stable internal environment
- b.
The mechanism by which sense organs communicate with the control center of the body
- c.
An internal and external structure of the body
- d.
The effective use of receptors in the body

11:46 PM



[Assistant](#)

a. The body's attempt to maintain relatively stable internal environment

Explanation: Homeostasis is the physiological process by which the body regulates internal conditions (temperature, pH, glucose levels, ion concentrations, etc.) within narrow limits despite external changes, using receptors, control centers, and effectors.

11:46 PM

What is the structural measurement in community studies that deals with the number of individuals per unit area?

Question 91Answer

a.

Diversity

b.

Species composition

c.

Population size

d.

Density

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d. Density

Explanation: Density is the number of individuals of a species per unit area (or volume) and is the structural measure used in community and population studies to quantify how crowded or sparse a population is.

11:48 PM

After conducting research and final reporting, uncovered and the remaining components the research can be mentioned in which section?

Question 92Answer

- a.
Statement of the problem
- b.
Results
- c.
Recommendations
- d.
References

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c. Recommendations

Explanation: After reporting results, any uncovered issues, remaining components, limitations, or suggestions for future work are typically presented in the Recommendations (or sometimes "Conclusions and Recommendations") section to guide further research or action.

11:48 PM

he largest gland in the human body is:

Question 93Answer

- a.
Pancreatic gland
- b.
Pituitary gland
- c.
Adrenal gland
- d.
Sweet gland

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None of the above — the largest gland in the human body is the liver.

Explanation: The liver is the largest gland and the largest internal organ. The options listed (pancreas, pituitary, adrenal, sweat glands) are much smaller in comparison.

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The largest gland in the human body is:

Question 93Answer

- a.
Pancreatic gland
- b.
Pituitary gland
- c.
Adrenal gland
- d.
Sweet gland

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Correct answer: None of the above — the largest gland in the human body is the liver.

Explanation: The liver is the largest gland and the largest internal organ. The given choices (pancreas, pituitary, adrenal, sweat gland) are all much smaller.

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Which of the following is NOT a requirement for commercially important fish?

Question 94Answer

- a.
Optimum pH and salinity
- b.
Optimum fertilizer
- c.
Light
- d.
Dissolved oxygen

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b. Optimum fertilizer

Explanation: Commercially important fish require suitable water conditions such as optimum pH and salinity (a), appropriate light (c) for some species' behavior/food production, and sufficient dissolved oxygen (d). "Optimum fertilizer" is not a direct requirement for fish themselves—fertilizers are sometimes used to stimulate primary production in aquaculture ponds, but fertilizer per se is not a universal requirement for commercially important fish.

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One is true about the difference between monocots and dicots.

Question 95Answer

- a.
Dicots have two embryo
- b.
Monocots do not have endosperm
- c.
Dicots have two cotyledons
- d.
Monocots and dicots have similar endosperm

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c. Dicots have two cotyledons

Explanation: Dicots (dicotyledons) characteristically have two cotyledons (seed leaves) in the embryo. Monocots have one cotyledon. The other options are false: (a) dicots have two cotyledons, not two embryos; (b) monocots typically do have endosperm (often prominent); (d) monocots and dicots differ in seed and endosperm patterns, so they are not necessarily similar.

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Which of the following alternative shows how the innate immunity is protecting the body against infection?

Question 96Answer

a.

Natural Killer cells kill infected host cells

b.

Neutrophils engulf bacteria

c.

T cells kill infected cells or help activate other immune cells

d.

Macrophages present germ parts to activate adaptive immunity

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b. Neutrophils engulf bacteria

Explanation: Neutrophils are innate immune cells that rapidly respond to infections by phagocytosing (engulfing) and destroying bacteria.

Notes on the other options:

- a. Natural killer (NK) cells are part of innate immunity and do kill infected or abnormal host cells, so this is also an innate mechanism—but among the choices, neutrophil phagocytosis is the classic, direct innate defense against bacteria.
- c. T cells are components of adaptive immunity (not innate).
- d. Macrophages are innate cells that can present antigen to activate adaptive immunity, but antigen presentation bridges innate and adaptive systems; the most straightforward, primary innate action listed is neutrophils engulfing bacteria.

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One of the following is true about the transmission modes of Taeniasis?

Question 97Answer

- a.
By eating contaminated fruits & vegetables
- b.
During blood donation
- c.
By eating raw or under cooked infected meat
- d.
By drinking contaminated water & food

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- c. By eating raw or undercooked infected meat

Explanation: Taeniasis (infection with adult tapeworms such as *Taenia saginata* or *Taenia solium*) is acquired by ingesting raw or undercooked beef or pork containing infective cysticerci. The other options describe transmission routes for different parasites or pathogens.

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The property that distinguishes Prokaryotic cells from Eukaryotic cells is the presence of?

Question 98Answer

a.
Membrane bounded organelles

b.
Cytoplasm

c.
A plasma membrane

d.
DNA

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a. Membrane bounded organelles

Explanation: Eukaryotic cells have membrane-bound organelles (such as a nucleus, mitochondria, endoplasmic reticulum), whereas prokaryotic cells lack these structures. Cytoplasm, a plasma membrane, and DNA are present in both prokaryotes and eukaryotes.

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Which organic compound contains one or more amino groups ($-\text{NH}_2$) and one or more carboxyl groups ($-\text{COOH}$)?

Question 99Answer

a.
Glycerol

b.
Glucose

c.
Fatty acid

d.
Amino acid

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d. Amino acid

Explanation: Amino acids contain at least one amino group ($-\text{NH}_2$) and one carboxyl group ($-\text{COOH}$). Glycerol is an alcohol, glucose is a monosaccharide, and fatty acids have a carboxyl group but no amino group.

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Which of the following process of cellular respiration occurs in cytoplasm of a cell?

Question 100Answer

a.

Krebs cycle

b.

Acetyl-CoA reaction

c.

Glycolysis

d.

Electron transport chain

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c. Glycolysis

Explanation: Glycolysis occurs in the cytoplasm and converts one glucose molecule into two pyruvate molecules, producing a net gain of 2 ATP and 2 NADH. The Krebs cycle and electron transport chain occur in the mitochondria (matrix and inner membrane, respectively), and the conversion of pyruvate to acetyl-CoA occurs in the mitochondrial matrix (or prokaryotic cytoplasm in bacteria).

